

# Follow-Up Analysis: Six Questions Answered

## Poets + Quran Clustering — Extended Results

### Q1 — Which surah was the outlier in Experiment 2?

**Surah 111: Al-Masad.** The only surah (out of 114) that didn't fit into any cluster.

### Q2 — Percentage by surah count vs. by text size

| Group                       | Surahs (count) | % by count | Total ayat | % by ayat |
|-----------------------------|----------------|------------|------------|-----------|
| Surah-only clusters         | 84             | 73.7%      | 5,389      | 86.4%     |
| Mixed clusters (with poets) | 29             | 25.4%      | 841        | 13.5%     |
| Outlier                     | 1              | 0.9%       | 5          | 0.1%      |

By text volume, the surah-only clusters look even more dominant (86.4% of all ayat) than by simple count (73.7%) — the surahs that mix with poets tend to be shorter than average. The picture doesn't change direction either way: most of the Quran's actual text groups separately from poets; a real but smaller minority does not.

### Q3 — Full cluster membership table

Every cluster's full poet and surah membership was exported to `output/tables/full_cluster_membership.csv` — 27 cluster rows across both experiments, with complete name lists. Too long to usefully print here; the cluster-size summary is below.

| Experiment | Cluster                     | Poets | Surahs    |
|------------|-----------------------------|-------|-----------|
| Exp 1      | Cluster 1 (Quran's cluster) | 19    | 1 (Quran) |
| Exp 1      | Outliers                    | 40    | 0         |
| Exp 1      | 11 other clusters           | 201   | 0         |
| Exp 2      | Cluster 0                   | 0     | 61        |
| Exp 2      | Cluster 1                   | 3     | 22        |
| Exp 2      | Cluster 6                   | 9     | 5         |
| Exp 2      | Cluster 3, 5 (surah-only)   | 0     | 23        |
| Exp 2      | Outliers                    | 10    | 1         |
| Exp 2      | 6 poet-only clusters        | 238   | 0         |

## Q4 — Do the 19 poets already cluster together, without the Quran?

Reclustered the 260 poets alone (no Quran at all, same method as the original paper). **18 of the 19 poets (95%) already land together**, split across just two adjacent clusters:

| Where they land (no Quran) | Poets | % of the 19 |
|----------------------------|-------|-------------|
| Cluster 2                  | 12    | 63%         |
| Cluster 3                  | 6     | 32%         |
| Outlier                    | 1     | 5%          |

**This is the most important new finding.** These 19 poets aren't an artifact of adding the Quran — they already formed a tight, pre-existing stylistic sub-group in the poetry-only analysis. The Quran's embedding happened to land inside (or next to) a cluster that already existed independently.

## Q5 — Rerun excluding those 19 poets (241 + Quran = 242)

| Metric                      | Original (261)          | Excluding the 19 (242)        |
|-----------------------------|-------------------------|-------------------------------|
| Clusters found              | 13                      | 12                            |
| Quran's status              | In Cluster 1 (19 poets) | Outlier (with 48 other poets) |
| Nearest-neighbor similarity | 0.971 - 0.976           | 0.969 - 0.972                 |

Without its 19 closest poets, the Quran no longer forms a clean cluster with anyone — it drops into the outlier bucket, alongside 48 other poets who also don't fit cleanly anywhere. But note: its raw similarity to its new nearest poets barely moved (0.969 vs. 0.976). So removing the 19 didn't make the Quran look meaningfully less similar to everyone else — it just wasn't quite dense enough on its own to register as a defined cluster under this method's threshold.

## Q6 — Rerun restricted to size-matched poets (213 poets + 114 surahs)

Restricted to the 213 of 260 poets (82%) whose entire surviving output is comparable in size to a single surah (3–286 verses total) — removing prolific, well-documented poets like Zuhayr, Tarafa, and Amr ibn Kulthum from the pool.

| Metric                   | Original<br>(374, all poets) | Size-matched<br>(327, 213 poets) |
|--------------------------|------------------------------|----------------------------------|
| Silhouette score         | 0.469                        | 0.607 (much cleaner)             |
| Surahs in mixed clusters | 30 / 114 (26%)               | 45 / 114 (39%)                   |
| Surahs as outliers       | 1 / 114                      | 0 / 114                          |

**This result argues for, not against, the earlier sample-size explanation.** Restricting to only minor poets didn't reduce the overlap with the Quran — it increased it, from 26% to 39% of surahs. That's the expected direction if minor poets (thin, less-distinctive style signatures) are what's driving the overlap: removing the well-documented, highly-distinctive major poets left a pool skewed toward exactly the kind of less-differentiated poet whose embedding tends to sit closer to everything, including the Quran.

## Updated Overall Read

Putting all six answers together: the 19-poet overlap from the first report is real and not a fluke of adding the Quran (Q4) — but it's also fragile (Q5, disappears without its core group) and gets stronger, not weaker, once major/prolific poets are excluded (Q6). That combination points toward a sample-size/text-volume effect being the more likely driver than a genuine deep stylistic resemblance to a specific handful of poets.

**This still measures writing-style similarity, not authorship.** Nothing in this follow-up changes that: these results describe measurable patterns in the embeddings, not evidence about who wrote what.